POLYNOMIAL PIECES FOR A PERFECT GENERATOR LIST ARE OPEN AND CLOSED SUBSTACKS

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 626-642 of sources/stacks-moduli/pinned-original-source.tex; proof span is lines 644-672, the complete author proof (27 body lines).

Editorial note. Navigation stubs: 5 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 13 of queue_moduli.json (pb=1170, stmt 626-642, proof 644-672).

Editorial note. stacks-moduli wave-1: fresh adjudication accept (2026-10-14); pool per Chief order 08:5x (dpa superseded to lane-s9); moduli.tex 53 labels/48 with proof judged 25 accept / 23 reject / 5 pb=0 env-dup rows excluded; spans cite-free and reference/slogan/history-free so no patchenv needed.

Editorial note. Author's 'we suggest the reader skip the proof' is tone; the Noetherian-reduction and perfect-descent steps are written out and complete.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 0DND: sources/stacks-moduli/pinned-original-source.tex, source lines 588–597. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0DNC: sources/stacks-moduli/pinned-original-source.tex, source lines 555–585. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 07SK: sources/stacks-moduli/pinned-original-source.tex, source lines 672–672. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0851: sources/stacks-moduli/pinned-original-source.tex, source lines 672–672. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Date: stacks-project:moduli.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 09RJ: sources/stacks-moduli/pinned-original-source.tex, source lines 672–672. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $f : X \to B$ be as in the introduction to this section. Let $E_1, \ldots, E_r \in D(\mathcal{O}_X)$ be perfect. Let $I = \mathbf{Z}^{\oplus r}$ and consider the map*

$$I \longrightarrow D(\mathcal{O}_X), \quad (n_1, \ldots, n_r) \longmapsto E_1^{\otimes n_1} \otimes \ldots \otimes E_r^{\otimes n_r}$$

Let $P : I \to \mathbf{Z}$ be a map. Then $\mathcal{C}oh^P_{X/B} \subset \mathcal{C}oh_{X/B}$ as defined in Situation 0DNC is an open and closed substack.

Proof. We may work étale locally on B, hence we may assume that B is affine. In this case we may perform absolute Noetherian reduction; we suggest the reader skip the proof. Namely, say $B = \operatorname{Spec}(\Lambda)$. Write $\Lambda = \operatorname{colim} \Lambda_i$ as a filtered colimit with each Λ_i of finite type over $\mathbf{Z}$. For some i we can find a morphism of algebraic spaces $X_i \to \operatorname{Spec}(\Lambda_i)$ which is separated and of finite presentation and whose base change to Λ is X. See Limits of Spaces, Lemmas 07SK and 0851. Then after increasing i we may assume there exist perfect objects $E_{1,i}, \ldots, E_{r,i}$ in $D(\mathcal{O}_{X_i})$ whose derived pullback to X are isomorphic to $E_1, \ldots, E_r$, see Derived Categories of Spaces, Lemma 09RJ. Clearly we have a cartesian square

$$\begin{array}{ccc} \mathcal{C}oh^P_{X/B} & \longrightarrow & \mathcal{C}oh_{X/B} \\ \downarrow & & \downarrow \\ \mathcal{C}oh^P_{X_i/\operatorname{Spec}(\Lambda_i)} & \longrightarrow & \mathcal{C}oh_{X_i/\operatorname{Spec}(\Lambda_i)} \end{array}$$

and hence we may appeal to Lemma 0DND to finish the proof. □